



Extraordinary MHNet: Military high-level camouflage object detection network and dataset

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<https://doi.org/10.1016/j.neucom.2023.126466> 

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Abstract

We present the first systematic work on Military High-level Camouflage object Detection (MHCD), aiming to identify objects visibly embedded in chaotic backgrounds. The high intrinsic similarities (e.g., texture, intensity, color, etc.) between the attention object and its background give the task far more challenging than general object detection. In this paper, we construct a benchmark MHCD2022 dataset, which consists of 3000 images with dense annotations covering 5 categories from multiple real-world scenes. Remarkably, based on the observation that biological vision usually first obtains perception from global search and strives to recover the complete object, we propose a novel Military High-level detection Network, called MHNet, which is characterized by four ingenious modules: Subject Perception Gathering (SPG), Part-object Relationships Mining (PRM), Concept Recovery/Feature Clue Supplement (CR/FCS) and Springboard Selection (SS). Firstly, a SPG is designed for global foreground rough perception by the exploitation of depth information. Second, a PRM is particularly used to mine part-object potential relations in diverse environments. After that, we propose CR/FCS and SS to enhance the destroyed instance-level representation and suppress the domain imbalance problem, respectively. Extensive experimental results show that previous methods suffered from poor performance, MHNet significantly outperforms camouflage baselines and competing methods on the MHCD2022 for the high-level camouflaged object. Finally, we also present and highlight the practical application value and several future directions of the research.

Introduction

Can you quickly find and locate the high-level camouflaged military objects from each image in Fig. 1? We define objects that are difficult for the human eye to immediately discern as high-level camouflage and those that can be quickly distinguished as the lower-level. Given an arbitrary image, the goal of general object detection is to determine predefined classes of presences and locate them in this image. In contrast, identifying concealed/ camouflaged objects in a biological vision scene becomes more intractable. Those objects are masters who are concealed in the surrounding environment. They have other appearances that imitate the same body colors, texture, pattern, and other backgrounds, such as dazzle camouflage [1], camouflage [2], disruptive coloration [3], and distractive markings [4]. This misleading information hinders the most discriminating feature of network learning camouflage. We believe that more visual perception knowledge/prior about camouflage objects is beneficial.

Recently, camouflage patterns have been widely used in military equipment, and the technique makes it harder to efficiently distinguish an object's appearance from a concealed background, as the foreground object has a very similar appearance to its surroundings. Visual cues used for identification such as texture, edges, color, contrast, and scale are vulnerable to attacks by basic camouflage strategies e.g. disruptive coloration. Closely related work brings convenience to the decipherment of high-level camouflage techniques for warfighter or UAV reconnaissance. To the best of our knowledge, military camouflage object detection has not been studied in the literature before, especially for high-level military camouflage patterns. One of the main reasons is that no accessible object detection benchmark has so far favored the military domain for high-level camouflage. As a result, previous detection methods with military objectives have been mainly grounded in conventional operations such as multi-sensor information processing [5] and migration learning [6], and readers have rarely seen works that directly use deep learning. From the perspective of visual attention mechanisms, as a close task, the relationship between saliency object detection (SOD) and camouflage recognition is orthogonal and adversarial, with SOD working to distinguish the most attention-grabbing regions rather than MHCD ought to reduce the initial saliency level of foreground objects, i.e. the game theory of conspicuous and unseen objectives. Here, it is clearly beyond the scope of this paper to provide an adequate review of the SOD, and we encourage the reader to refer to recent research work [7], [8] for the desired details. The suitability of the above approach is tenuous in the concealed context, we aim to explore prior contributions for high-level camouflaged object detection. Moreover, other hardworking researchers have proposed some common methods for infrared and visible image fusion [50], [51], [52], [53]. In particular, Tang L et al. [50] proposed a semantic-aware real-time image fusion network (SeAFusion) by bridging the gap between image fusion and high-level vision tasks. SuperFusion [52] incorporates image registration, image fusion, and semantic requirements of high-level vision tasks into a single framework, and achieves satisfactory experimental results. Visible and infrared fusion approaches will be the focus of our future research efforts due to the reconciliation of the mutual advantages and development potential of both. As one of the cores of vision fundamentals, we only discuss

visible-image-based approaches for deciphering tasks at this stage in this paper and will not repeat them here.

Based on the observation of biological vision, predators initially perceive “abruptness” (i.e., inconsistency) from a global scene with patterns almost identical to the object and pop up possible rough regions when searching for camouflaged prey; Following this, the predator carefully analyzes those regions in which possible objects and potential parts of the body are based on the a priori knowledge learned from previous stages; further treatments are similar to the instances-level active recovery or optimization of the camouflaged objects. As such, those camouflaged objects can be correctly discriminated against. However, the perfect treatment of the MHCD problem faces several pitfalls as follows: (1) the actual military environment is more complex and the object scale is more variable. (2) Blurred background. (3) High intrinsic similarities of textures between the body and the background. (4) Intra-class differences, same-class objects such as persons with specific patterns (e.g. weed cloak or snowsuit) may be very different. (5) Inter-class similarity, objects belonging to different classes may be very similar, e.g. tanks and military vehicles. In general, military high-level camouflage patterns are usually focused on more cluttered environments such as battlefields, jungles, towns, or mountains. The lighting and obscuration situations are extremely complex and harsh. Therefore, it is difficult to be identified by general detection methods.

To contribute to the development of this research area, we propose a novel MHNet framework and construct a dataset in an attempt to deal with the problem of discrimination in the military context, under high-level camouflage scenes. MHNet consists of four intuitive modules: Subject Perception Gathering (SPG), Part-object Relationships Mining (PRM), Concept Recovery/Feature Clue Supplement (CR/FCS), and Springboard Selection (SS). To capture the global “abrupt” information representation, SPG is specially designed to generate a depth feature architecture similar to the geoscientific “contour topography” by using shallow features containing rich boundary information and detail information. Inspired by promising results [9], we took the initiative to introduce such a process of part and object potential property mining into MHCD, considering the aspect that most previous methods ignore that an object normally consists of several related parts combined, and related parts can form the whole object. Next, we propose a CR/FCS component with reverse attention for rescuing or recovering cues interrupted by obstacles such as occlusion or local feature presentation, which is similar to the “brain patching” process in human behavior. In addition, to cope with the problem of multi-source data, where inter-domain inconsistency of samples is analyzed, we propose a simple and efficient SS module to stabilize the model in the case of hard and easy samples with an unfavorable skew towards the latter, which tends to cause the network to fail in learning hard samples. The main contributions of this paper are listed as follows:

- (1) Based on careful bio-visual observation, we propose a novel Military High-level detection Network, called MHNet, for military high-level camouflage object detection. It will facilitate the development of individual combat systems on the real

battlefield to achieve rapid object detection and pre-emptive strike, which is a favorable basis for motivating new ideas on this task.

- (2) We design a global "abrupt" perceptual strategy SPG module to obtain subject-aware a priori by constructing a depth representation for objects, which can discover the agnostic coarse location of the object and ameliorate the weak boundary problem.
 - (3) We propose a PRM module that integrates part-object relations based on the knowledge of SPG and deep feature encoding learning to enhance the capture of the whole object by the model with the supplementary part. To our knowledge, this is the first preliminary attempt to incorporate part-object relationship mining into military high-level camouflage object detection.
 - (4) To mimic human brain activity, we further propose a concept recovery strategy, termed CR/FCS, to repair the destroyed features by fusing reverse attention search with boundary features. In addition, the SS method is cleverly designed to alleviate the domain inconsistency problem brought by training on multiple sources of data and successfully forces the model to take into account the stable learning of "hard, easy" samples.
 - (5) We created a solid benchmark, the Military High-level Camouflage Dataset MHCD2022, to contribute significantly to the future research process. Through the experiments in the MHCD2022, the proposed MHNet method greatly outperforms several advanced baselines and has significant advantages and friendliness in the identification of high-level camouflaged objects compared with the state-of-the-art methods.
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Section snippets

Related work

Types of Camouflage. Camouflaged samples can be roughly divided into two types, i.e. natural camouflage and artificial camouflage. Natural camouflage is used by animals as survival skills, such as seahorses, starfish, and chameleons, to confuse predators. Furthermore, artificial camouflage is often used in commercial designs/games to hide information. It appears in life products during the manufacturing process, or in military equipment (the kind we care about).

Generic Object Detection (GOD). ...

Datasets

Datasets play a pivotal role in learning-based object detection methods. Unfortunately, to our knowledge, evaluation works on the existence of military camouflage object detection are rare and informal. To date still, no datasets designated for military camouflage and especially for high-level concealment grand-master are available. Thus, this has inspired our determination to promote this work. Domain-specific datasets are necessary because the characteristics of the objects present large gaps ...

MHNet

In this section, we first illustrate the proposed overall architecture of MHNet as shown in Fig. 2. The four key components are respectively elaborated in detail. ...

Datasets and Implementation Details

We evaluate the performance of the proposed method on the military camouflage dataset MHCD. Our whole dataset is divided into a training set and a test set according to a 4-to-1 ratio. 2400 images are expanded to 12,000 images by classical image reversal, rotation, offset, and mosaic enhancement algorithms. Thus, 12,000 images are finally used for training and the remaining 600 images are used for testing. The specified categories and the number of instances are presented in Table 1. In ...

Conclusion and Future work

This paper investigates for the first time high-level military concealed object detection. Specifically, we have provided the new challenging and carefully labeled MHCD2022 benchmark and developed a novel but efficient end-to-end perception and identification framework (i.e. MHNet) based on human visual mechanisms. Our MHNet achieves robust and superior results when tackling objects with a high degree of concealment compared to other works that exist. The above contributions have provided the ...

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

Acknowledgments

This work was partially supported by the Natural Science Foundation of Heilongjiang Province of China under Grant No.LH2021F026; the Fundamental Research Funds for Central Universities under Grant No.HIT.NSRIF202243. ...

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